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Long-term functional recovery after first ischemic stroke: The Northern Manhattan Study

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Abstract

Background and Purpose—Several factors predict functional status after stroke, but most studies have included hospitalized patients with limited follow-up. We hypothesized that ischemic stroke patients experience functional decline over 5 years independent of recurrent stroke and other risk factors.

Methods—In the population-based Northern Manhattan Study, incident ischemic stroke patients ≥ 40 years were prospectively followed using the Barthel index (BI) at 6 months and annually to 5 years. Baseline stroke severity was categorized as mild (NIH Stroke Scale < 6), moderate (6–13), and severe (≥ 14). Follow-up was censored at death, recurrent stroke, or myocardial infarction. Generalized Estimating Equations provided odds ratios (OR) and 95% confidence intervals (95%CI) for predictors of favorable (BI ≥ 95) versus unfavorable (BI < 95) functional status, after adjusting for demographic and medical risk factors.

Results—Of 525 patients, mean age was 68.6 ± 12.4 years, 45.5% were male, 54.7% Hispanic, 54.7% had Medicaid/no insurance, and 35.1% had moderate stroke. The proportion with BI ≥ 95 declined over time (OR 0.91, 95% CI 0.84–0.99). Changes in BI by insurance status were confirmed by a significant interaction term (β for interaction = -0.167 , $p = 0.034$); those with Medicaid/no insurance declined (OR 0.84, $p = 0.003$), whereas those with Medicare/private insurance did not (OR 0.99, $p = 0.92$).

Conclusions—The proportion of patients with functional independence after stroke declines annually for up to 5 years, and these effects are greatest for those with Medicaid or no health insurance. This decline is independent of age, stroke severity, and other predictors of functional decline, and occurs even among those without recurrent stroke or myocardial infarction.

Keywords

disability; stroke; recovery

Introduction

Stroke is the leading cause of serious, long-term disability in the United States.¹ Considering the staggering prevalence of stroke,² the burden of post-stroke disability is of primary public health importance. Prior research on the natural history of disability after stroke has shown varying degrees of functional recovery within 6–12 months.^{3–6} Long-term functional outcomes are less clear, however.

Previous studies have identified demographic and stroke-specific characteristics that are associated with functional recovery after stroke.^{3–5, 7–14} However, most of these studies have included only hospitalized patients with limited follow-up, and few population-based studies have examined predictors of functional status with long-term follow-up. Furthermore, the effect of socioeconomic status on functional status after stroke has not been examined over the long-term.¹⁵

We sought, in a prospective, population-based, multi-ethnic, urban stroke cohort study, to determine the long-term natural history and predictors of functional status among participants who experienced a first ischemic stroke. We hypothesized that ischemic stroke patients experience functional decline over 5 years independent of recurrent stroke and other risk factors.

Subjects and Methods

The Northern Manhattan Study (NOMAS) includes a population-based incident ischemic stroke follow-up study designed to determine predictors of stroke recurrence and prognosis in a multi-ethnic, urban population, as previously described.^{16, 17} The race-ethnic mixture of the northern Manhattan community at the time of the study consisted of 63% Hispanic, 20% black, and 15% white residents.

Selection of NOMAS cohort

The methods of patient identification and enrollment have been described in previous publications.^{18, 19} Briefly, ischemic stroke patients were enrolled if they: (1) were diagnosed with a first stroke; (2) were age ≥ 40 years, and (3) resided in Northern Manhattan for ≥ 3 months in a household with a telephone. NOMAS was limited to ischemic stroke to restrict cases to a more homogeneous population and limit the underlying pathophysiologies. Case ascertainment occurred between July 1993 and June 1996, and assessments were completed in August 2001. Over 80% of patients with acute ischemic stroke in northern Manhattan are hospitalized at the Columbia University Medical Center (CUMC). Subjects hospitalized at other local hospitals were identified through active surveillance of admissions to those hospitals and through agreements with local physicians. Approximately 5% of incident ischemic stroke patients in northern Manhattan were not hospitalized and were also included.^{16, 19} Evaluation of patients was performed at the hospital; those subjects either not hospitalized or hospitalized elsewhere were evaluated in the outpatient research clinic. Of the participants who were not hospitalized at CUMC, only those who had an initial evaluation within 20 days of their first stroke were included in analyses involving the NIH Stroke Scale. The study was approved by the CUMC Institutional Review Board. All participants gave consent directly or through a surrogate when appropriate.

Index Evaluation of Subjects

Data were collected through interviews by trained research assistants, and physical and neurological examinations were conducted by study neurologists, as previously described.^{18, 19} Assessments were conducted in English or Spanish depending upon the primary language of the participant. Race-ethnicity was based upon self-identification. Standardized questions were adapted from the Behavioral Risk Factor Surveillance System²⁰ by the Centers for Disease Control and Prevention regarding the following conditions: hypertension (HTN), diabetes mellitus (DM), hypercholesterolemia, peripheral vascular disease, transient ischemic attack, cigarette smoking, and cardiac conditions such as myocardial infarction (MI), coronary artery disease, angina, congestive heart failure (CHF), atrial fibrillation, other arrhythmias, and valvular heart disease. HTN was defined as a systolic blood pressure recording ≥ 160 mmHg or a diastolic blood pressure recording ≥ 95 mm Hg (based on the average of two blood pressure measurements) or the patient's self-report of a history of HTN or antihypertensive use. DM was defined by a fasting blood glucose level ≥ 126 mg/dL, the participant's self-report of such history, or insulin or oral hypoglycemic use.^{18, 19} Presence or absence of urinary incontinence within 7–10 days of the index stroke was determined. Presence or absence of depressed mood in the week prior to stroke was assessed at enrollment. Physical activity was assessed using a questionnaire adapted from the National Health Interview Survey of the National Center for Health Statistics, and classified as light-moderate and heavy, as previously described.¹⁸

Stroke severity was assessed using the NIHSS score derived from a standardized neurological examination, and was categorized into mild (<6), moderate (6–13), and severe (≥ 14). This categorization was based on previous analyses of stroke severity in relation to stroke outcome from our population and a clinical trial.^{21, 22} Stroke diagnostic evaluation included computerized tomography and/or magnetic resonance imaging of the brain, ultrasound evaluation and/or magnetic resonance angiography of the extracranial and intracranial cerebral vessels, and transthoracic or transesophageal echocardiogram as appropriate. A consensus of stroke neurologists assessed stroke subtype using modified Stroke Data Bank criteria and all available information, as described in a previous publication.²³ Stroke neurologists also determined brain side of index stroke, presence of aphasia, and parietal lobe dysfunction including neglect.

Follow-Up and Outcomes Assessment

Follow-up evaluations were conducted at six months and then annually for five years. Evaluations were conducted at the medical center or by telephone interaction with the patient, family member, or caregiver. There was also a telephone assessment at 1.5 years. Information on vital status, functional status as measured by the Barthel index (BI), and intercurrent symptoms, illness, or hospitalization was collected. Previous research has demonstrated the reliability of phone assessments of functional status using the BI.²⁴ In-person assessments included interviews and assessment of functional status, as well as measurement of vital signs, physical and neurological examination.

Patients unable or unwilling to come to the medical center were visited by a member of the research staff, and the evaluation was conducted at home or in an alternative place of residence (e.g., nursing home). An ongoing surveillance system of admissions to the CUMC and other local hospitals, described in a previous publication,¹⁶ was also used to identify study participants who experienced recurrent stroke, MI, hospitalization, or death. When available, medical records were reviewed for all outcome events including death. All outcome events were initially reviewed by a specially-trained research assistant, and events were then adjudicated by study neurologists or cardiologists, as appropriate.

Statistical Analyses

Statistical analyses were conducted using SAS Version 9.1.3 (SAS Institute, Cary, NC). For descriptive purposes, means were calculated for continuous variables and proportions for categorical variables. Follow-up was censored at the time of death, recurrent stroke, or MI because of the effect of these events on functional status. We used Generalized Estimating Equations (GEE) to assess the relationship between several predictor variables, including time of follow-up at regular intervals, with functional outcome. The advantage of GEE is that it accounts for within-subject correlation and produces unbiased and more efficient regression estimates in studies that have a longitudinal, repeated-measures design.²⁵ GEE with binary outcome and logistic link function provided parameter estimates, from which odds ratios and 95% confidence intervals (OR, 95% CI) were calculated for predictors of favorable (BI $\geq$ 95) versus unfavorable (BI <95) functional status, after adjusting for demographic (age, sex, race/ethnicity, education level, marital status, insurance status) and medical (HTN, coronary artery disease or MI [CAD], DM, hypercholesterolemia, smoking, physical activity) risk factors, as well as stroke characteristics known to be associated with functional recovery in previous studies (initial stroke severity, depressed mood prior to stroke, presence or absence of aphasia, presence of parietal lobe dysfunction including neglect, side of stroke, and presence of urinary incontinence). Insurance status was characterized as Medicare/private insurance versus Medicaid/no insurance. Multivariate models were constructed in a backward stepwise manner using those variables that were significant at $p < 0.10$. In secondary analyses, we evaluated predictors for BI $\geq$ 60 versus <60.

Subgroup analysis was performed in those who had BI $\geq$ 95 at 6 months, in order to determine the natural history and predictors of functional status in those deemed “recovered” after the acute phase of recovery.

The interaction between time of the follow-up assessment and insurance status was included in multivariate regression models to assess whether the change in functional status over time differed by insurance status. After testing for interaction with insurance, the effect of follow-up time on functional status was also stratified by type of insurance.

Time of follow-up assessment was analyzed both continuously and discretely (at 1, 1.5, 2, 3, 4, and 5 year time points, with 0.5 year as reference). The analysis of discrete time points did not assume any particular functional form, whereas the continuous model assumed a linear time trend. Using the model with discrete time points, we tested whether the trend was linear. In addition, in the continuous model, appropriateness of linearity was determined by testing whether the quadratic term was significantly different from zero.

To explore whether there was a significant decrement in functional status at a single time point -- a change point -- maximum likelihood estimates of the change point were obtained. To assess whether differential length of follow-up was related to functional status, which would have been a source of bias, a series of logistic regression models was fitted for each time point using an indicator of whether each subject dropped out after each follow-up time point as the outcome, and BI $\geq$ 95 and the same covariates in our final model as the predictors.

Results

There were 655 patients enrolled in the NOMAS stroke incidence and follow-up study. No follow-up functional data was available for participants who died within 6 months ($n = 83$) and for those who died between 6 months and 1 year ($n = 14$). Five participants died after 1 year, but before functional data were collected. Twenty-four participants had recurrent stroke before first functional assessment and were excluded from analysis. There were no functional data and no indication of death for 4 participants, who were categorized as lost to follow-up. Five

hundred twenty-five patients were thus available for this analysis. In addition, during follow up, 84 patients had recurrent stroke or MI, and there were 133 deaths. Table 1 lists baseline and demographic information for study participants.

There was an annual decline in the proportion of patients with $BI \geq 95$ in an unadjusted regression model considering time to follow-up assessment as a continuous variable (OR per year after stroke 0.96, 95% CI 0.92–1.01). When adjusted for demographic variables, the annual decline was significant (adjusted OR 0.94, 95% CI 0.89–1.00) and did not change further when adjusted for demographic variables and medical risk factors (adjusted OR 0.94, 95% CI 0.88–0.99). Further adjusting for stroke characteristics, such as stroke severity, neglect, side of stroke, and urinary incontinence, led to a slightly more pronounced decline (adjusted OR per year after stroke 0.91, 95% CI 0.84–0.99). Results for the cutoff of $BI \geq 60$ showed more marked declines; an unadjusted model showed OR of 0.93 (95% CI 0.87–0.98), and models adjusted for demographic variables (adjusted OR 0.90, 95% CI 0.84–0.96) and demographics and medical risk factors (adjusted OR 0.89, 95% CI 0.83–0.96) showed that with every year of follow-up, participants were less likely to have $BI \geq 60$. Table 2 reports the fully adjusted results.

In subgroup analysis for those participants who had $BI \geq 95$ at 6 months ($n=245$), there was still a significant annual decline in the proportion with $BI \geq 95$ over follow-up (unadjusted OR per year 0.81, $p<0.0001$), and there was little change when adjusted for demographic variables (adjusted OR 0.79, $p<0.0001$) and in the fully adjusted model (adjusted OR per year 0.76, 95% CI 0.65–0.89; Table 3).

Changes in BI over time differed significantly by insurance status (β for interaction = -0.167 , $p=0.034$), with a decline in BI over time for those with Medicaid/no insurance (adjusted OR 0.84, 95% CI 0.75–0.94, $p=0.003$), but no definite decline among those with Medicare/private insurance (adjusted OR 0.99, 95% CI 0.90–1.11, $p=0.92$). No other two-way interactions between time to follow-up and other variables were significant. Figures 1 and 2 depict functional status stratified by insurance status, unadjusted for other risk factors.

In the final regression model, time was treated as a continuous variable in linear form because there was no evidence that an alternative model better fit the data (lack of linearity Chi-square test: $df=4$, $p=0.65$).

In order to evaluate at which time point the greatest change occurs, we conducted an analysis of the change point in functional decline. The maximum likelihood estimate of the change point was at 3 years, and the change in proportion of participants with $BI \geq 95$ before and after the 3 year time point was significantly different from zero (OR 0.76, $p=0.0088$), indicating that the greatest functional decline began to occur 3 years after stroke.

There was no significant difference in functional status when those with shorter follow-up were compared to those with relatively longer follow-up, confirming that differential follow-up does not depend on functional status.

Other predictors of $BI \geq 95$ among all participants were age at stroke, diabetes, marital status, stroke severity, and baseline urinary continence (Table 2). Results were similar for the BI cutoff of 60, with the exception that DM approached but did not reach significance as a predictor ($p=0.0842$) and marital status was not associated with outcome ($p=0.94$). Left-sided stroke (versus right) was associated with better functional outcome. Among participants with $BI \geq 95$ at 6 months, predictors were similar to those among all participants, except that Hispanic ethnicity was a significant predictor whereas severe stroke was not (Table 3).

Discussion

In this large, population-based, prospective cohort study of first-time ischemic stroke patients, we observed a modest annual decline in functional status over the long term after stroke, even after censoring those with recurrent stroke and MI and adjusting for risk factors and baseline stroke severity. There was a significant decline over years of follow-up even among those with early functional “recovery” (BI $\geq$ 95 at 6 months). In the United States, there are 4 tiers of insurance access: private insurance (69% of the U.S. population in 2004²⁶), Medicaid (a state-administered insurance only available to low income residents that has been associated with limited access to health care in New York State^{27, 28}; 13% of the population), Medicare (a federally-administered insurance primarily for residents aged $\geq$ 65 years; 14% of the population), and no insurance (16% of population). Functional decline in our study was seen particularly among those who were uninsured or insured with Medicaid. This decline began to be apparent at about 3 years when we used as a threshold for disability the BI threshold of 95, and at 2 years with a BI cutoff of 60.

Few community-based studies have examined long-term disability after stroke. One study reported that 37% of first stroke patients worsened in Rankin scores between 1 and 5 years after stroke in Rochester, Minnesota.²⁹ Limitations of this study included comparison of mean Rankin scores without reporting of confidence intervals and the lack of multivariable analysis for predictors of outcomes. Furthermore, the GEE analysis that was used in our study is more sensitive to individual changes in disability over time. In another study, dependence in ADL tasks in 109 first stroke patients was analyzed, and a smaller percentage of women were independent at 5 years and there was a greater proportion of dependence among older patients.³⁰ However, this study had a small sample, reported only percentages in each category without confidence intervals or significance tests, did not conduct multivariable analysis, and used a potentially insensitive ADL measurement scale. Another study examined modified BI at 5 years in 129 survivors of first stroke and found that 36% developed new major disability, with predictors including older age, drowsiness at baseline assessment, moderate hemiparesis, and recurrent stroke.³¹ This study was limited by small numbers of data available at 5 years and only one follow-up assessment of disability. Finally, none of these studies censored recurrent stroke or cardiac events, and hence it is unclear whether worsened disability over time was the result of recurrent stroke or long-term effects of a single index stroke. In contrast to these studies, our study collected data at multiple timepoints and excluded those with recurrent clinical vascular events and so was able to assess a linear change in function over time.

The delayed functional decline that we observed has not been well-described before and has several possible explanations. Non-stroke comorbidities that are known to affect functional status may contribute in the long-term, as prior research in non-stroke cohorts has shown.³² In particular, cognitive dysfunction has been shown to be associated with functional status after stroke.^{33, 34} In cognitive research, the concept of “cognitive reserve” refers to differing susceptibility to cognitive impairment that is related to variables such as education, literacy, intelligence quotient, and engagement in leisure activities.³⁵ Considering the close correlation between cognitive status and functional status, there may be a similar phenomenon with regard to performing ADLs that we may term “functional reserve.” The deficit caused by a first stroke may result in a depleted functional reserve and a consequent failure to compensate for brain aging. We were unable to include detailed measurements of cognitive function in our analyses, however.

Another possible explanation for the observed delay in functional decline among uninsured and Medicaid patients is that the stroke caused a functional deficit that affected participants equally regardless of insurance status. Stroke, in other words, may serve as a sort of equalizer of function due to its direct biological effects on the brain, as well as the availability of acute

rehabilitation and other therapies available to participants with all types of insurance. Over time after stroke, however, functional status among those with private insurance and Medicare and those with Medicaid or no insurance may diverge, due to disparities in care and more limited access to rehabilitative services, information about health, and ongoing management of risk factors and chronic conditions^{28, 36} that are known to have an impact on functional status.³⁷

The delayed decline in functional status could also be due to the ceiling effect that has been observed with the BI.³⁸ It is possible that there was a steady decline in functional status that began soon after ischemic stroke, but since the BI is insensitive to small changes in disability, this decline was not captured until 3 years after the stroke.

Another possible explanation for the observed decline in functional status is that participants may have experienced clinically silent recurrent strokes during follow-up. Studies have shown a prevalence of 6–18% of clinically silent strokes in different populations depending on risk factors and imaging protocols,³⁹ and silent infarcts may be as much as 5 times as prevalent as symptomatic infarcts.⁴⁰ A previous study in the Northern Manhattan population showed that 18% of 892 participants free of clinical stroke had subclinical infarcts.⁴¹ Despite the term “silent” stroke, these subclinical events may affect functional status.⁴²

Previous studies with limited follow-up have shown predictors of functional status similar to the predictors that we found, in particular age,^{3, 4, 7, 8} DM,^{4, 8, 9} marital status (a marker of social support),^{5, 10} stroke severity,^{4, 7, 8, 11} side of stroke,^{11, 12} and urinary continence.^{13, 14} Most previous studies, however, have assessed these functional outcomes at single points in time, whereas in our analysis we were able to assess the associations for these risk factors with functional outcomes measured at multiple annual intervals. These risk factors thus continue to predict functional status over several years. The fact that these predictors were found to be significant in our study suggests that they represent robust associations with functional status, for selection bias is minimized in population-based studies and follow-up continued to 5 years post-stroke. The fact that DM was a significant predictor of functional outcome is notable considering the high prevalence of DM in this population, which is likely due to a combination of environmental (including dietary) and genetic factors.

In our multi-ethnic population-based study, race/ethnicity was not associated with functional outcome among all participants, in univariate or multivariate analyses. Socioeconomic status and access to care were primary mediating effects of functional status, perhaps mitigating the effect of race/ethnicity on outcome, which has been previously described.⁴³

Limitations of this study include use of the BI to assess disability instead of a disability scale designed specifically for stroke patients. However, the BI is widely used in stroke research, which allows comparison of this study with prior studies. Second, detailed data on rehabilitation received after the stroke was not collected, and may have informed the relationship between late functional decline and insurance status. Third, since some of the data on baseline risk factors was obtained by self-report, uninsured participants who do not regularly get medical care may not be aware of undiagnosed conditions. Hence, in order to lessen bias, we used laboratory and blood pressure measurements and information about prescription medications to capture risk factors such as DM, HTN, and hypercholesterolemia. Fourth, neuropsychological data were not analyzed in this study, which could have informed the link between cognitive status and functional status. Future research will be directed towards clarifying the relationship between time and functional decline, comparing functional status over time in stroke patients and non-stroke controls matched for risk factors, and assessing the role of cognitive decline as a mediator of the change in functional status. Finally, the decline in functional status among those with less access to care highlights the need to improve the

ways in which knowledge of risk factor control is translated in clinical practice to improve outcome following stroke.

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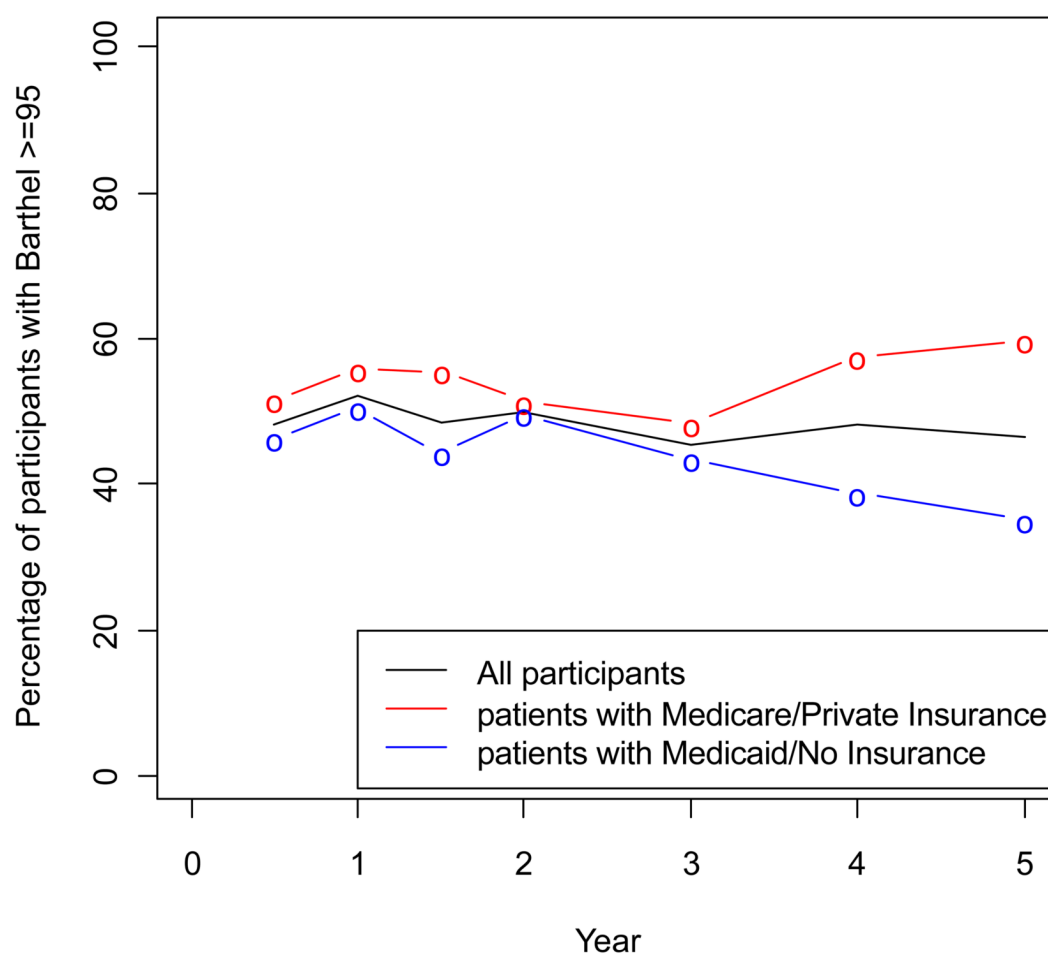


Figure 1.
Percentage of participants with Barthel ≥ 95 , stratified by insurance status.

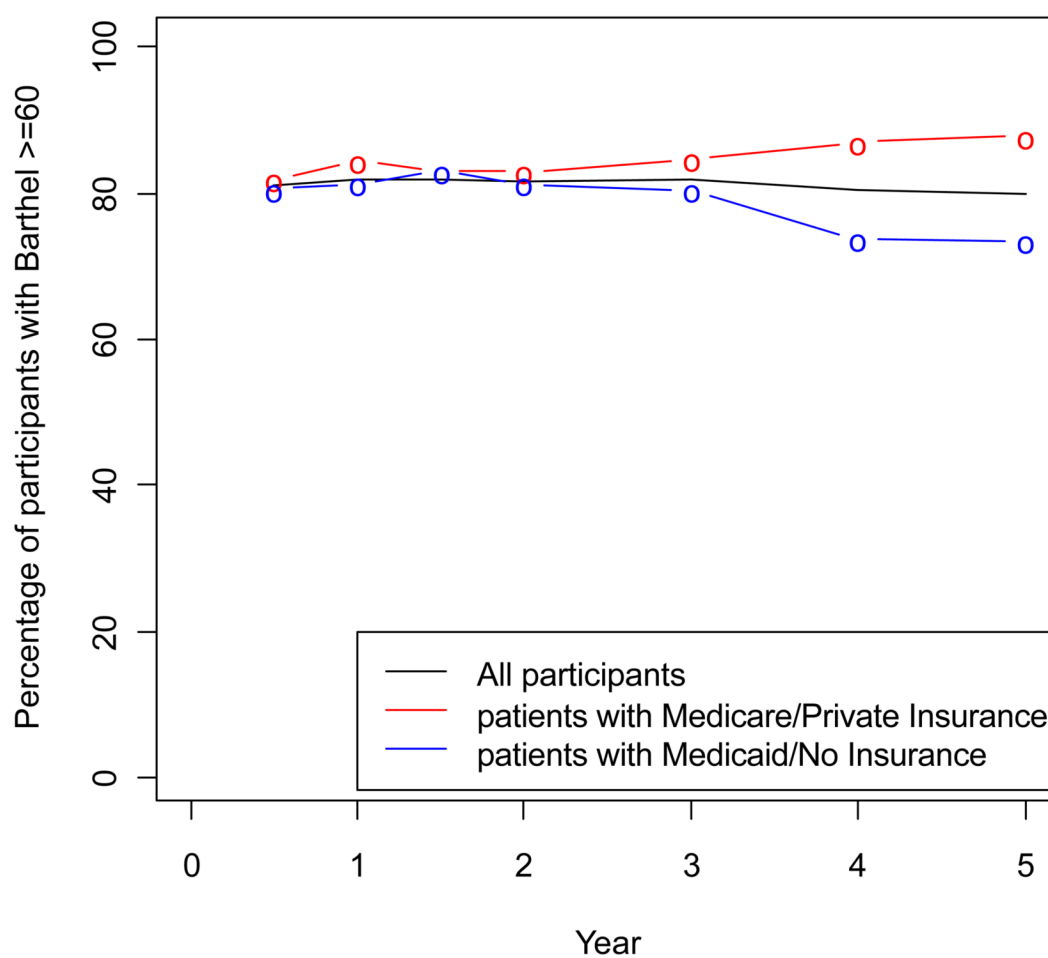


Figure 2.
Percentage of participants with Barthel ≥ 60 , stratified by insurance status.

Table 1

Baseline characteristics of study population*

Number of participants (n)[†]	525
Demographics:	
Age, mean (SD), y	68.6 (12.4)
Male, No. (%)	239 (45.5)
Received at least high school education, No. (%) (n=513)	169 (32.9)
Marital status, No. (%) married (n=517)	178 (34.4)
Non-Hispanic White, No. (%)	85 (16.2)
Non-Hispanic Black, No. (%)	141 (26.9)
Hispanic, No. (%)	287 (54.7)
Other race, No. (%)	12 (2.3)
Insured with Medicaid or uninsured, No. (%)	279 (54.7)
Insured with Medicare or private insurance, No. (%)	231 (45.3)
Risk factors, No. (%) (n=524)	
History of myocardial infarction	87 (16.6)
History of coronary artery disease	156 (29.7)
History of atrial fibrillation	51 (9.7)
History of peripheral arterial disease	114 (21.8)
Current smoking	113 (21.6)
Past smoking	177 (33.8)
Diabetes mellitus	236 (45.0)
Hypertension	443 (84.4)
Hypercholesterolemia	277 (52.8)
Depressed mood prior to stroke (n=511)	160 (31.3)
Stroke etiologic subtypes, No. (%)	
Atherosclerotic	79 (15.1)
Lacunar	135 (25.7)
Cardioembolic	86 (16.4)
Cryptogenic	210 (40.0)
Stroke severity, No. (%) (n=431)	
NIHSS rating 0–5	234 (54.6)
NIHSS rating 6–13	152 (35.1)
NIHSS rating ≥14	45 (10.3)
Side of stroke, No. (%) right-sided (n=520)	293 (56.4)
Incontinent of urine 7–10 days after stroke (n=481)	139 (28.9)

* SD=standard deviation; NIHSS=National Institutes of Health Stroke Scale; definitions of risk factors and stroke characteristics in text.

[†] all percentages calculated with n=525 unless otherwise noted.

Table 2
Multivariate generalized estimating equation (GEE) regression model of predictors of function after ischemic stroke*

Variable	For Barthel ≥ 95 vs. < 95			For Barthel ≥ 60 vs. < 60		
	Odds ratio (OR)	95 % confidence interval of OR	p-value	Odds ratio (OR)	95 % confidence interval of OR	p-value
Time of follow-up assessment	0.91	0.84–0.99	0.019	0.87	0.78–0.97	0.015
Age at stroke	0.95	0.93–0.96	<.0001	0.93	0.90–0.95	<.0001
Male sex	1.41	0.89–2.22	0.14	1.34	0.75–2.38	0.32
Race-ethnicity [†]						
Black	0.88	0.42–1.86	0.74	1.80	0.75–4.31	0.19
Hispanic	1.05	0.50–2.17	0.91	1.53	0.66–3.51	0.32
At least high school education	1.64	0.94–2.87	0.082	1.72	0.89–3.32	0.11
Insured with Medicaid or uninsured	0.85	0.53–1.37	0.51	0.90	0.49–1.66	0.74
Physical activity	1.49	0.99–2.24	0.057	1.68	0.98–2.89	0.060
DM	0.48	0.32–0.73	0.0006	0.61	0.35–1.07	0.084
Married status	1.82	1.12–2.97	0.016	1.02	0.54–1.95	0.94
CAD	0.72	0.47–1.09	0.12	1.29	0.71–2.37	0.41
Moderate stroke [‡]	0.36	0.24–0.55	<.0001	0.26	0.14–0.48	<.0001

Variable	For Barthel ≥ 95 vs. < 95				For Barthel ≥ 60 vs. < 60			
	Odds ratio (OR)	95 % confidence interval of OR	p-value		Odds ratio (OR)	95 % confidence interval of OR	p-value	
Severe stroke [‡]	0.06	0.02–0.18	<.0001		0.02	0.01–0.05	<.0001	
Urinary continence	3.80	2.27–6.36	<.0001		3.32	1.83–6.04	<.0001	
Left sided stroke	1.48	0.99–2.22	0.059		1.90	1.07–3.36	0.027	

* DM=diabetes mellitus; CAD=coronary artery disease or myocardial infarction

[†] compared to non-Hispanic white race-ethnicity

[‡] compared to mild stroke

Table 3

Multivariate generalized estimating equation (GEE) regression model of predictors of function after ischemic stroke, among those with BI ≥ 95 at 6 months*

Variable	Odds ratio (OR)	95 % confidence interval of OR	p-value
Time of follow-up assessment	0.76	0.65–0.89	0.0005
Age at stroke	0.95	0.91–0.98	0.004
Male sex	1.25	0.65–2.41	0.50
Race-ethnicity [†]			
Black	0.50	0.11–2.32	0.37
Hispanic	0.21	0.05–0.89	0.03
At least high school education	2.16	0.88–5.29	0.09
Insured with Medicaid or uninsured	0.97	0.48–1.98	0.94
Physical activity	1.33	0.69–2.56	0.39
DM	0.47	0.23–0.92	0.03
Married status	2.47	1.11–5.49	0.03
CAD	0.58	0.29–1.16	0.12
Moderate stroke	0.33	0.16–0.67	0.002
Severe stroke	0.17	0.01–2.17	0.17
Urinary continence	5.42	2.18–13.50	0.0003
Left sided stroke	1.08	0.58–2.04	0.80

* BI=Barthel index; DM=diabetes mellitus; CAD=coronary artery disease or myocardial infarction

[†] compared to non-Hispanic white race-ethnicity